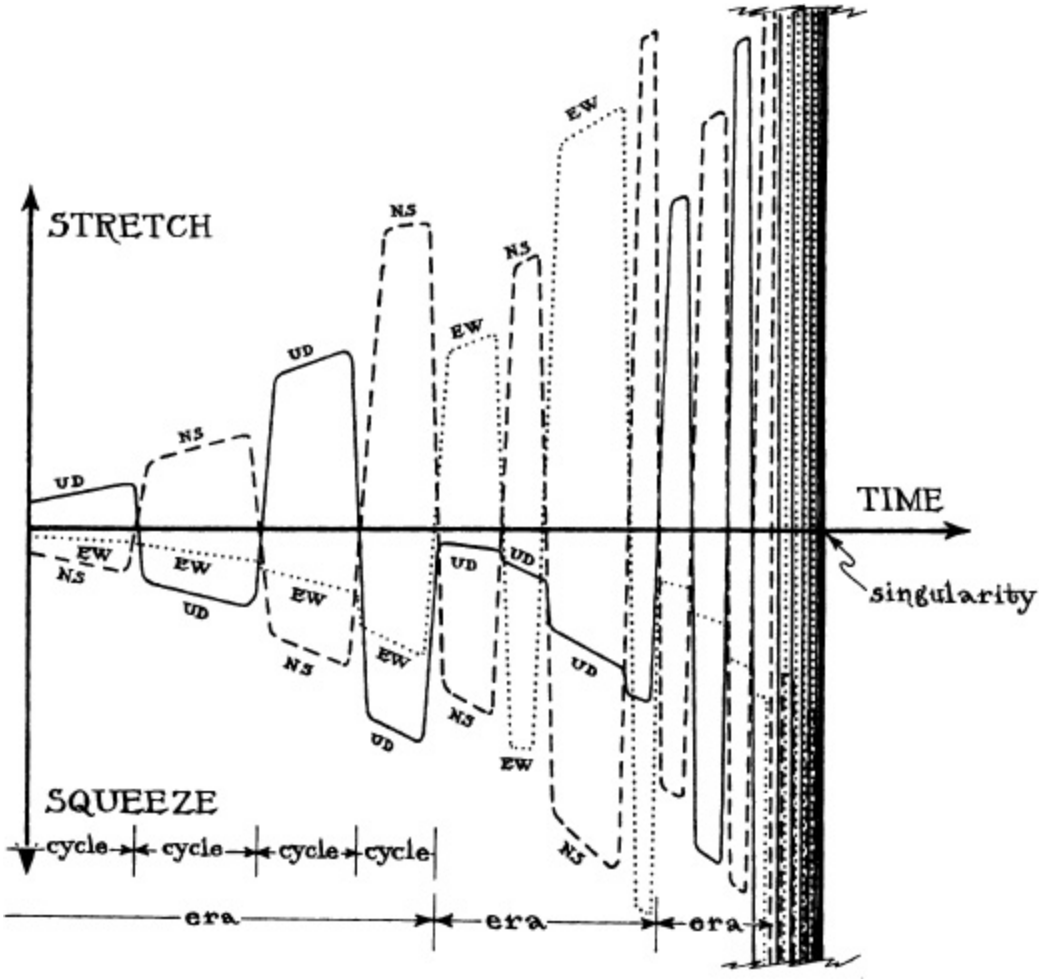
example of how the tidal forces might oscillate, but the precise sequence of oscillations is chaotically unpredictable.

In Misner's version of the mixmaster singularity, the oscillations were the same everywhere in space, at a particular moment of time (as measured, say, by the astronaut). Not so for the BKL singularity. Its oscillations are spatially chaotic as well as temporally chaotic, just as turbulent motions of the froth in a breaking ocean wave are chaotic in space as well as in time. For example, while the astronaut's head is being alternately stretched and squeezed ("pummeled") along the north/south direction, his right foot might be pummeled along the northeast/southwest direction, and his left foot along south-southeast/north-northwest; and the frequencies of oscillation of the pummeling might be quite different on his head, his left foot, and his right foot.



13.6 An example of how the tidal forces might oscillate with time in a BKL singularity. The tidal forces act in different manners along three different, perpendicular directions. These